

KIMHYONGGWON, Korea, Democratic People's Republic ofWMO: **470220**Lat: **40.817N**Lon: **128.150E**Elev: **3937**StdP: **12.72**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.3	-10.8	-20.0	2.1	-12.5	-16.9	2.5	-9.4	27.3	7.6	20.8	-0.6	3.0	320	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	79.9	67.2	77.2	66.1	74.7	64.5	70.5	77.1	68.6	74.8	66.9	72.4	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.3	120.7	75.0	66.5	113.0	72.7	64.9	106.7	70.7	37.2	77.8	35.4	74.9	33.9	72.7	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	16.4	13.3	-19.9	85.2	4.0	3.1	-22.7	87.4	-25.0	89.2	-27.3	90.9	-30.2	93.2		
(5)			-19.5	73.1	4.0	2.7	-22.4	75.0	-24.8	76.6	-27.1	78.1	-30.0	80.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	6.6	12.2	24.4	38.7	51.0	58.5	65.1	64.0	53.8	41.9	25.5	10.9
	DBStd	21.64	7.84	8.59	9.32	8.14	6.54	4.53	4.59	5.02	5.49	6.46	9.44	8.53
	HDD50	5840	1345	1057	793	345	66	2	0	0	26	259	736	1211
	HDD65	10015	1810	1477	1258	788	437	197	56	78	336	716	1186	1676
	CDD50	1414	0	0	0	8	96	259	467	435	141	8	0	0
	CDD65	110	0	0	0	0	2	4	57	47	0	0	0	0
	CDH74	707	0	0	0	3	55	90	338	216	5	0	0	0
	CDH80	89	0	0	0	0	7	5	51	26	0	0	0	0
Wind	WSAvg	3.6	4.0	4.3	4.8	4.7	4.8	2.8	2.2	2.1	2.6	3.2	3.7	3.9
Precipitation	PrecAvg	26.7	0.3	0.4	0.6	1.3	2.7	4.1	6.5	5.5	2.9	1.1	0.9	0.5
	PrecMax	43.6	1.3	2.4	1.4	2.7	13.3	6.6	13.4	11.5	8.0	2.7	2.9	1.9
	PrecMin	19.5	0.0	0.0	0.1	0.3	0.7	1.8	2.8	1.0	0.4	0.0	0.0	0.0
	PrecStd	6.5	0.2	0.5	0.3	0.6	2.3	1.5	2.9	2.5	2.0	0.7	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	42.7	56.5	71.0	79.3	80.4	85.3	83.3	74.3	67.4	53.7	38.5
		MCWB	26.2	36.5	43.6	50.4	59.9	62.5	70.5	72.2	61.4	52.2	43.7	32.8
	2%	DB	27.3	36.9	49.5	63.8	74.4	76.7	80.5	79.0	70.9	62.7	48.8	33.2
		MCWB	23.0	30.7	39.0	47.9	56.5	61.5	69.0	68.8	59.5	49.4	40.1	28.9
	5%	DB	23.8	32.3	44.3	59.1	70.1	73.3	77.5	76.1	68.2	58.8	43.9	29.0
		MCWB	20.3	27.1	35.6	45.6	53.8	60.1	67.9	67.8	57.9	47.2	37.6	25.5
	10%	DB	20.5	28.1	39.8	54.6	66.0	70.1	74.9	73.3	65.3	54.7	39.8	25.1
		MCWB	17.6	23.8	33.1	42.8	52.3	59.1	66.2	66.3	56.5	45.3	34.9	22.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	36.5	44.9	53.7	62.2	66.9	73.3	73.9	65.1	54.6	45.0	34.1
		MCDB	31.0	40.0	54.9	64.7	74.7	75.5	81.5	81.7	70.3	64.0	50.5	37.5
	2%	WB	23.3	31.1	40.1	49.3	58.4	64.2	70.9	70.7	62.5	51.4	41.6	29.2
		MCDB	26.8	36.2	47.8	60.6	71.2	72.0	78.1	76.6	68.1	59.3	47.5	32.8
	5%	WB	20.6	27.2	36.4	46.3	56.0	62.3	69.2	68.7	60.0	48.8	38.3	25.6
		MCDB	23.6	32.0	43.7	56.8	67.0	70.0	75.9	74.4	65.5	56.6	43.3	29.0
	10%	WB	17.6	23.9	33.1	43.7	53.7	60.4	67.5	67.1	57.9	46.5	35.2	22.3
		MCDB	20.4	28.0	39.4	53.4	63.8	68.2	73.5	72.4	63.2	53.2	39.6	24.9
Mean Daily Temperature Range	5% DB	MDBR	23.0	23.4	21.4	22.3	23.4	20.8	17.0	16.9	21.6	24.0	21.4	21.9
		MCDBR	26.2	28.0	27.0	30.9	32.4	27.8	22.1	20.5	25.6	30.5	26.8	25.6
		MCWBR	22.5	22.2	18.4	18.2	16.9	14.5	12.6	12.3	14.9	18.0	19.1	21.4
	5% WB	MCDBR	25.9	26.8	26.0	28.7	28.5	23.1	19.2	18.0	20.2	26.9	24.9	24.9
		MCWBR	22.4	21.7	18.7	18.6	16.7	14.0	11.9	11.9	13.5	17.3	18.3	21.1
Clear-Sky Solar Irradiance	taub	0.311	0.367	0.440	0.507	0.536	0.564	0.507	0.461	0.408	0.401	0.351	0.311	
	taud	2.140	1.998	1.864	1.785	1.780	1.777	2.008	2.130	2.206	2.099	2.145	2.173	
	Ebn at Noon	260	259	252	244	241	234	247	255	261	245	242	249	
	Edh at Noon	33	45	58	67	69	69	55	47	41	40	33	30	
All-Sky Solar Radiation	RadAvg	787	1091	1428	1604	1713	1602	1452	1434	1345	1084	755	666	
	RadStd	42	62	85	107	157	151	169	123	128	87	95	41	